Retail Data Analysis: Omni-channel Customer Behavior&Sales Performance

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1. Data Loading and Merging

```
# Load required libraries
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.3
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.3
```

```
library(lubridate)
```

```
##
## Attaching package: 'lubridate'
```

```
## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

```
# Load datasets
customer <- read.csv("customer.csv")
product <- read.csv("product.csv")
store <- read.csv("store.csv")
```

```
## Warning in read.table(file = file, header = header, sep = sep, quote = quote, :
## incomplete final line found by readTableHeader on 'store.csv'
```

```
transaction <- read.csv("transaction.csv")

# Merge transaction data with store, customer, and product details
full_data <- transaction %>%
  left_join(store, by = c("store_id" = "id")) %>%
  left_join(customer, by = c("customer_id" = "id")) %>%
  left_join(product, by = c("product_id" = "id"))

# Convert created_at to POSIXct datetime format
full_data$created_at <- as.POSIXct(full_data$created_at,
                                   format="%Y-%m-%d %H:%M:%S")

# Preview the combined dataset
head(full_data)
```

```
##       id store_id customer_id product_id quantity  total          created_at
## 1 626490        1      421833         50        1    884 2018-11-30 18:49:07
## 2 111599        2      125389         49       40  32720 2018-05-25 06:35:06
## 3 383397        2      192320         38       87 157557 2018-10-16 11:08:36
## 4 371545        1      337829         38       38  68818 2018-10-11 11:47:29
## 5 351387        2      267599         39       93 175584 2018-10-03 01:34:23
## 6 282369        1      178099         38       72 130392 2018-09-04 20:11:17
##           type      city   email gender price
## 1 Ofline store     Depok Hotmail Female   884
## 2 Online store     Depok   Gmail Female   818
## 3 Online store     Depok   Gmail Female  1811
## 4 Ofline store     Depok   Gmail   Male  1811
## 5 Online store     Depok   Gmail Female  1888
## 6 Ofline store Tangerang Hotmail   Male  1811
```

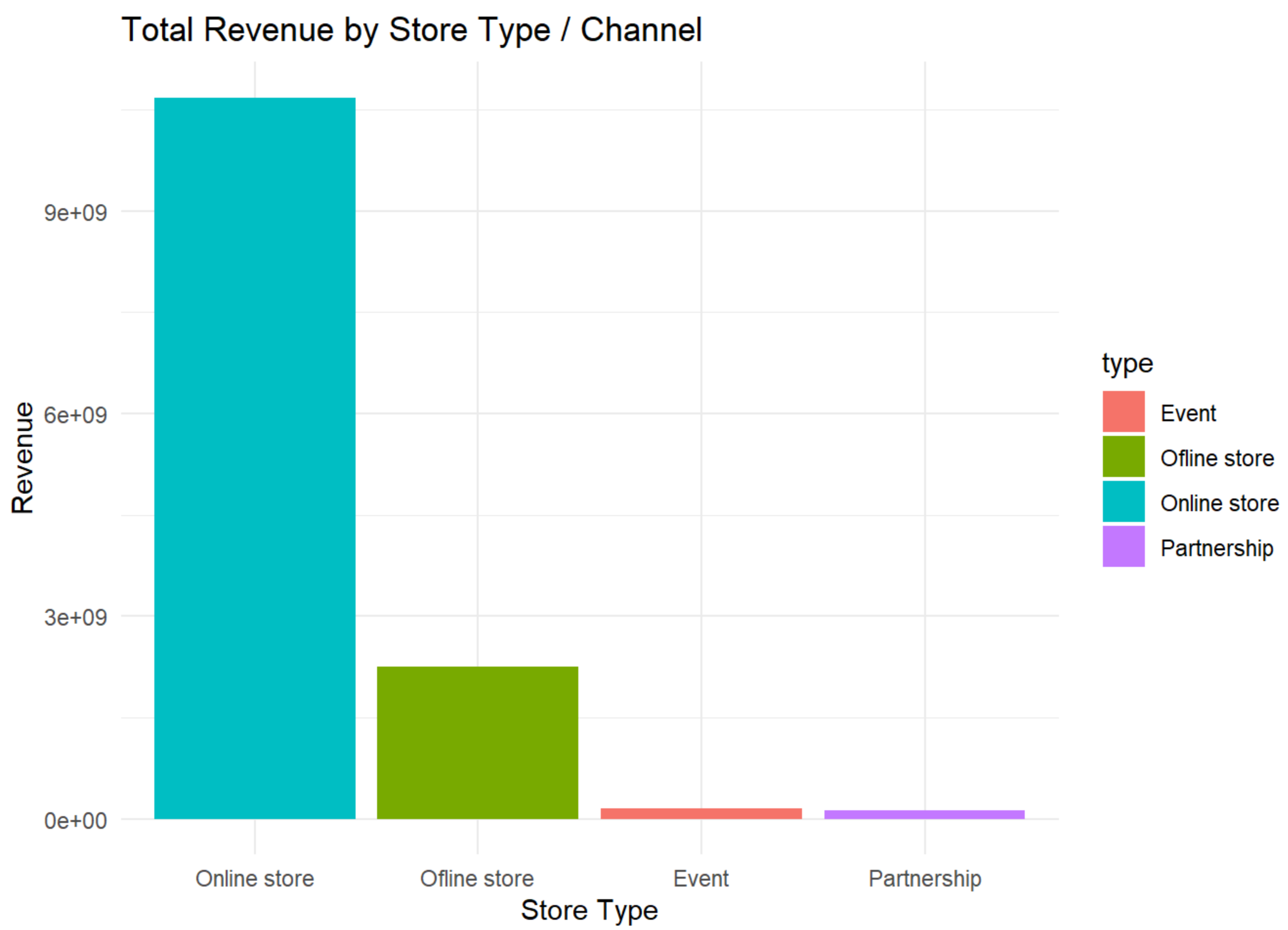
2. Channel Revenue and Performance Analysis

```
# Aggregating sales and quantities by store type
store_performance <- full_data %>%
  group_by(type) %>%
  summarize(
    total_revenue = sum(total, na.rm = TRUE),
    total_quantity = sum(quantity, na.rm = TRUE),
    transaction_count = n()
  ) %>%
  arrange(desc(total_revenue))

print(store_performance)
```

```
## # A tibble: 4 × 4
##   type         total_revenue total_quantity transaction_count
##   <chr>                <dbl>          <int>             <int>
## 1 Online store   10677116143        7372977            145764
## 2 Ofline store    2255852090        1591418             31693
## 3 Event            155038855         109879              2153
## 4 Partnership      123496360          90516              1777
```

```
# Visualization of Revenue by Store Type
ggplot(store_performance, aes(x = reorder(type, -total_revenue),
                              y = total_revenue, fill = type)) +
  geom_bar(stat = "identity") +
  theme_minimal() +
  labs(title = "Total Revenue by Store Type / Channel",
       x = "Store Type", y = "Revenue")
```



3. Customer Demographic Profiling

```r
# Spending behavior by gender and city
demographic_analysis <- full_data %>%
  group_by(city, gender) %>%
  summarize(
    avg_spending = mean(total, na.rm = TRUE),
    total_spending = sum(total, na.rm = TRUE)
  ) %>%
  arrange(desc(total_spending))
```

```
## `summarise()` has regrouped the output.
## ℹ Summaries were computed grouped by city and gender.
## ℹ Output is grouped by city.
## ℹ Use `summarise(.groups = "drop_last")` to silence this message.
## ℹ Use `summarise(.by = c(city, gender))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```r
print(demographic_analysis)
```

```
## # A tibble: 8 × 4
## # Groups:   city [4]
##   city      gender avg_spending total_spending
##   <chr>     <chr>         <dbl>          <dbl>
## 1 Depok     Female       73115.     2672351220
## 2 Depok     Male         72594.     2665363020
## 3 Jakarta   Male         73421.     1336926289
## 4 Tangerang Male         72226.     1322742818
## 5 Tangerang Female       73299.     1322165434
## 6 Bogor     Male         72933.     1298578801
## 7 Jakarta   Female       72660.     1298207574
## 8 Bogor     Female       72404.     1295168292
```

```r
# Preferred channel by gender
gender_channel <- full_data %>%
  group_by(gender, type) %>%
  summarize(count = n()) %>%
  mutate(percentage = count / sum(count) * 100)
```

```
## `summarise()` has regrouped the output.
## ℹ Summaries were computed grouped by gender and type.
## ℹ Output is grouped by gender.
## ℹ Use `summarise(.groups = "drop_last")` to silence this message.
## ℹ Use `summarise(.by = c(gender, type))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```r
print(gender_channel)
```

```
## # A tibble: 8 × 4
## # Groups:   gender [2]
##   gender type         count percentage
##   <chr>  <chr>        <int>      <dbl>
## 1 Female Event         1109      1.23
## 2 Female Ofline store 16042     17.8
## 3 Female Online store 72359     80.1
## 4 Female Partnership    833      0.922
## 5 Male   Event         1044      1.15
## 6 Male   Ofline store 15651     17.2
## 7 Male   Online store 73405     80.6
## 8 Male   Partnership    944      1.04
```

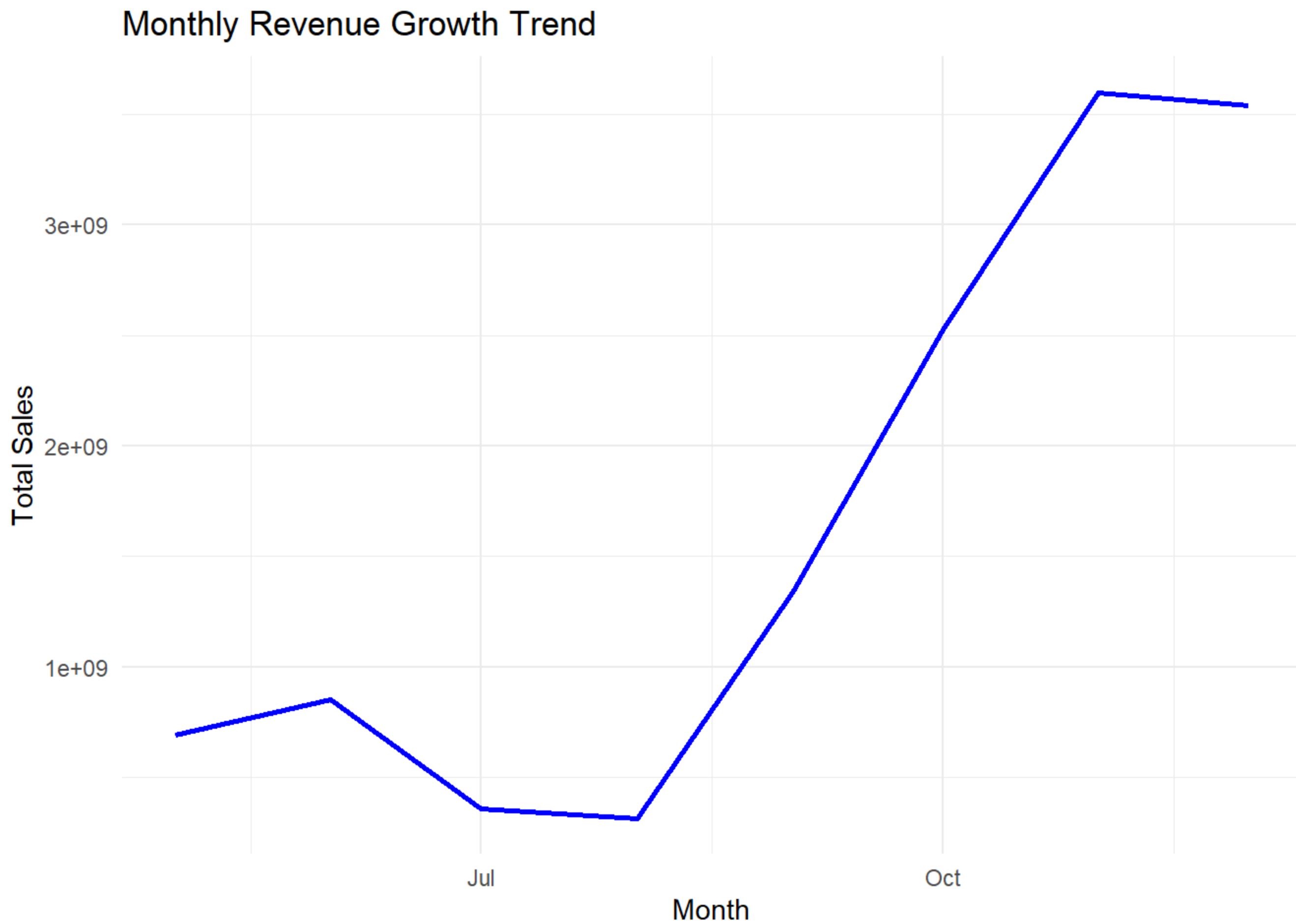
4. Time-Series / Seasonality Analysis

```
# Extracting month and hour of transaction
full_data <- full_data %>%
  mutate(
    month = floor_date(created_at, "month"),
    hour = hour(created_at)
  )

# Monthly Sales Trend
monthly_sales <- full_data %>%
  group_by(month) %>%
  summarize(monthly_revenue = sum(total, na.rm = TRUE))

ggplot(monthly_sales, aes(x = month, y = monthly_revenue)) +
  geom_line(color = "blue", size = 1) +
  theme_minimal() +
  labs(title = "Monthly Revenue Growth Trend", x = "Month", y = "Total Sales")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## ℹ Please use `linewidth` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Monthly Revenue Growth Trend
3e+09
2e+09
1e+09
Total Sales
Jul
Oct
Month